

The empirical recurrence for OEIS A227121, proved by transfer matrix

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Abstract

OEIS A227121 carries an empirical recurrence of order 5 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The entry's condition is local to a set of source states – those a source array consistent with the output emitted so far can be left in, one such state being a row of the source together with, for each adjacent pair of its columns, whether that pair is still equal or has already gone strictly less – sets with the same future behaviour being identified, so the admissible configurations of such a window are the vertices of a finite digraph and the arrays counted are exactly the walks in it; hence $a(n)$ is a walk count on $S = 7$ vertices and is C -finite. Whether the conjectured recurrence annihilates it is then settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A227121 is “Number of $n \times 2 \times 0,1$ arrays indicating 2×2 subblocks of some larger $(n+1) \times 3$ binary array having a sum of zero, with rows and columns of the latter in lexicographically nondecreasing order.”. It has offset 1 and begins

$$3, 7, 13, 23, 40, 68, 112, 178, \dots$$

The entry states, as a conjecture contributed by its author and never marked settled:

$$\text{Empirical: } a(n) = (1/24)n^4 - (1/12)n^3 + (11/24)n^2 + (31/12)n. \quad a(n) = 5a(n-1) - 10a(n-2) + 10a(n-3) - 5a(n-4) + a(n-5) \text{ for } n > 5.$$

As of the “Last modified” line on the live entry (Sep 07 2018 08:39:09, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a walk count

The entry counts distinct IMAGES and not the objects that produce them, so the vertices are not single configurations but sets of them: a set of source states – those a source array consistent with the output emitted so far can be left in, one such state being a row of the source together with, for each adjacent pair of its columns, whether that pair is still equal or has already gone strictly less – sets with the same future behaviour being identified. The set a word leaves the model in is determined by the word, and an edge from u to v carries the symbol emitted in passing, so distinct output words and walks from the initial vertex are the same thing. The condition itself is decided within the model by inspection of a state and the symbol read. An

admissible array is then exactly a walk, and ι, τ are the vectors recording which windows may begin and end an array, so with M the adjacency matrix on $S = 7$ vertices,

$$a(n) = \iota^\top M^n \tau,$$

the exponent fixed by the entry's own shape and checked against its published terms. In particular a is C -finite.

3 The proof

Lemma 1. *Let $q(t) = t^5 - \sum_i c_i t^{5-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M) \tau$ for the corresponding walk index j , so the recurrence holds from some point on if and only if $\iota^\top M^j q(M) \tau = 0$ for all large j , and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+5} - \sum_i c_i M^{j+5-i}$. For the bound, M^S is an integer combination of $I, M, \dots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M ; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. $\square$

Theorem 2.

$$a(n) = 5a(n-1) - 10a(n-2) + 10a(n-3) - 5a(n-4) + a(n-5)$$

for every $n > 5$.

Proof. The vector $w = q(M)\tau$ was formed by 5 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \dots$, again exactly; those integers vanish from the index corresponding to $n = 5 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. $\square$

4 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 41 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

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- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.